The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

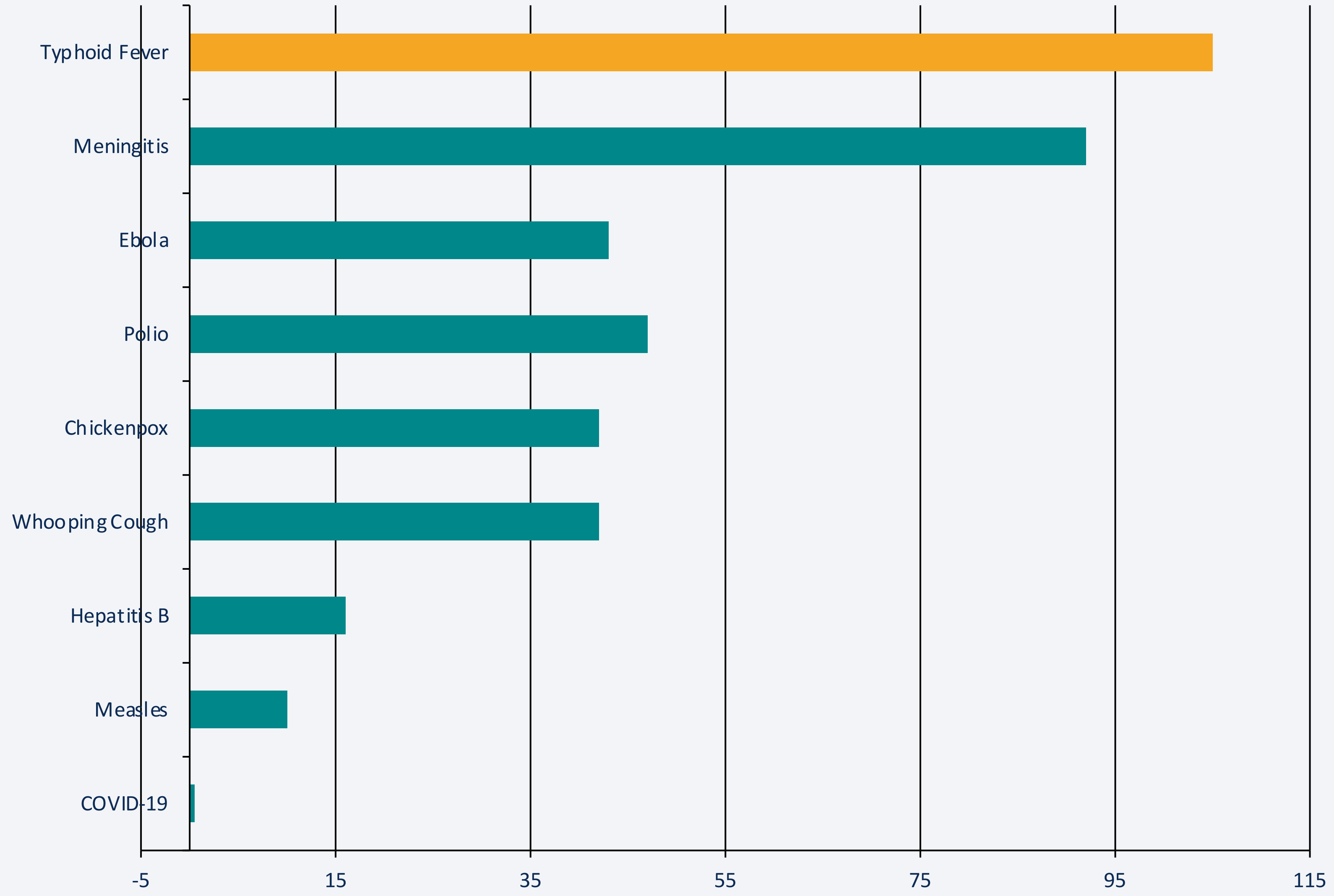
A data-driven overview · 1796 – 2020

Prepared for Public Health Professionals & Policy Analysts

Source: Our World in Data | Roush & Murphy (2007)

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen identification to vaccine licensure



Key Insight

Typhoid Fever

Identified 1884 → Licensed 1989

105-year gap

COVID-19

Identified Jan 2020 → Licensed Dec 2020

< 1 year — a record

Enabled by:

- mRNA technology
- Genome sequencing
- Advanced microscopy
- Virus-like particles

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre-vaccine vs. post-vaccine annual cases & deaths per million | Source: Roush & Murphy (2007)

Disease	Pre-Vaccine Cases/M/yr	Post-Vaccine Cases/M/yr	Case Reduction	Pre-Vaccine Deaths/M/yr	Death Reduction
Diphtheria	158	0	**100%**	13.7	**100%**
Smallpox	250	0	**100%**	2.9	**100%**
Acute Polio	141	0	**100%**	10	**100%**
Measles	3,044	0.2	**99.99%**	2.5	**100%**
Rubella	242	0.04	**99.98%**	0.09	**100%**
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute Hep. B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	**40.5%**	24	**31.3%**

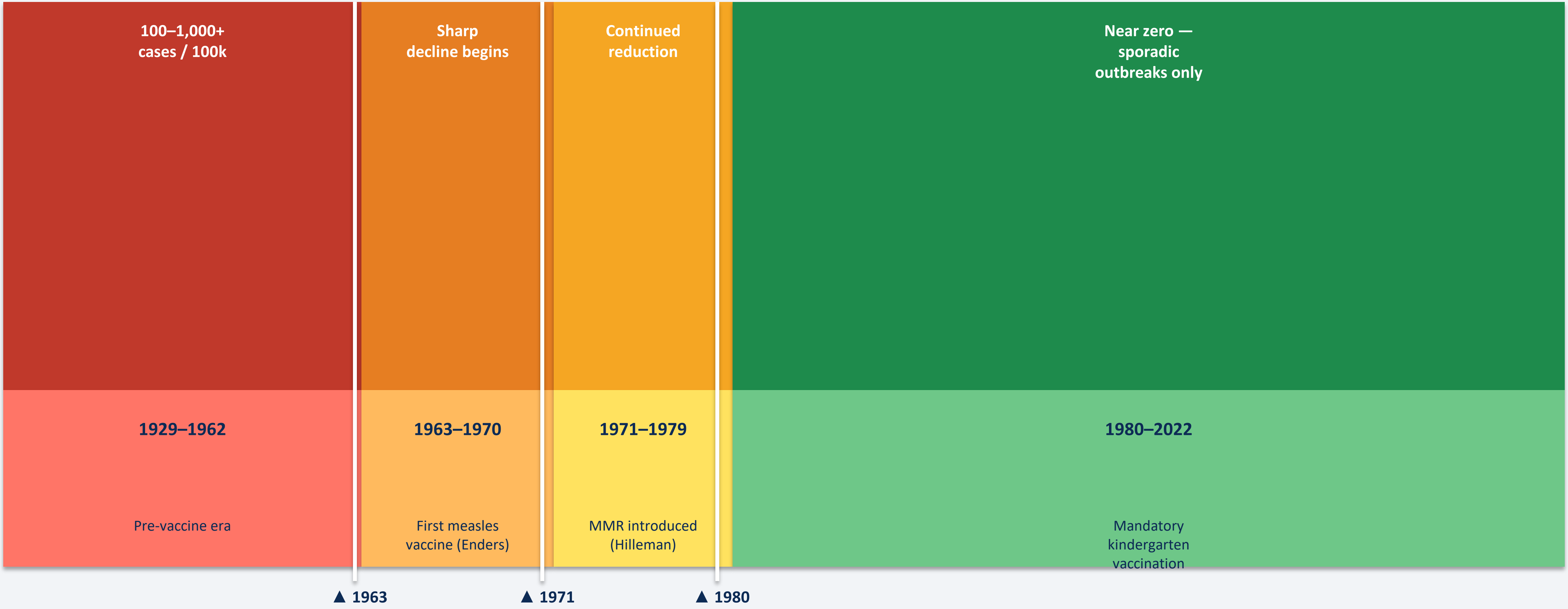
100% elimination

Incomplete control — ongoing gap

Source: Our World in Data · Roush & Murphy (2007)

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles cases per 100,000 by state, 1929–2022



Incidence scale (cases/100k): High (1000+) | Moderate | Low | Near Zero

Source: Our World in Data | Milestone annotations from source brief

Smallpox: The Only Human Disease Eradicated Through Vaccination

From WHO resolution to global certification | 1959 – 1980

1796 — Edward Jenner creates first smallpox vaccine

1959 — WHO passes eradication resolution

1967 — WHO intensified global campaign launched

1977 — Last natural case (Somalia)

1980 — WHO certifies global eradication — disease-free

Humans Only

No animal reservoir — virus could not hide in wildlife populations

Visible Symptoms

Distinctive rash made cases easy to identify and isolate quickly

Effective Vaccine

Cheap, heat-stable vaccine with lasting immunity after one dose

Global Will

WHO coordinated 59 countries in a sustained campaign

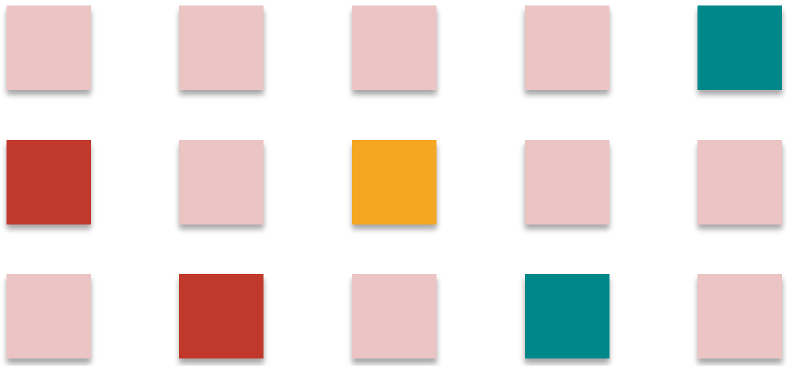
Source: Our World in Data · Roush & Murphy (2007)

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Population-level vaccination shields those who cannot be vaccinated

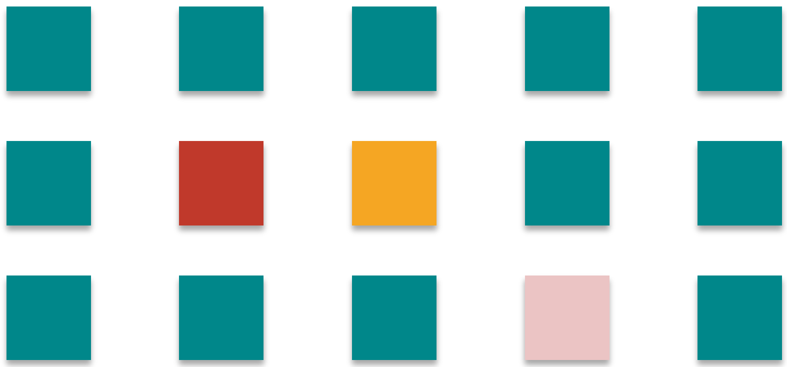
- Vaccinated
- Unvaccinated
- Vulnerable (can't vaccinate)
- Infected / spreading

LOW Coverage — Pathogen Spreads



→ Vulnerable person REACHED by pathogen

HIGH Coverage — Pathogen Cannot Spread (Herd Immunity)



→ Vulnerable person SHIELDED by surrounding immunity

Who Herd Immunity Protects

Newborns

Too young for many vaccines;
rely entirely on community immunity

Immunocompromised

HIV+ patients, organ transplant
recipients cannot mount
full immune response

Cancer Patients

Chemotherapy suppresses
immune system — vaccines
may be contraindicated

Elderly

Weakened immune response
means vaccines may
offer partial protection

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete disease control and systemic barriers to global vaccination

Disease	Case Reduction	Death Reduction
Pneumococcal	40.5%	31.3%
Hepatitis A	89%	88.7%
Acute Hep. B	83.9%	83.6%
Pertussis	96.6%	99.7%

vs. 100% reduction achieved for diphtheria, smallpox, and polio

Pneumococcal disease: 40.5% case reduction vs. 100% for diphtheria, polio, and smallpox.
Hepatitis A: 89% | Hepatitis B: 83.9%
Improved access, coverage, and vaccine formulations are urgently needed.

Access & Supply

Global vaccine supply shortages leave children in low-income countries at risk of preventable disease.

Scientific Trust

Poor public understanding and misinformation reduce vaccine uptake even where vaccines are available.

Incomplete Vaccines

Some vaccines (pneumococcal, hepatitis) provide only partial protection, requiring improved formulations.

Source: Our World in Data · Roush & Murphy (2007)

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

1

Vaccines achieved 100% elimination

Diphtheria, smallpox, and both forms of polio have zero cases and zero deaths in the US.

2

Development timelines collapsed

From 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.

3

Measles cases reached near zero

Across all 50 US states following the 1980 kindergarten vaccination mandate.

4

Herd immunity protects entire communities

Population-level vaccination shields newborns, immunocompromised patients, and those on chemotherapy.

5

Coverage gaps remain — action required

Pneumococcal (40.5%), Hepatitis A (89%), and Hepatitis B (83.9%) need improved access, coverage, and trust.

Call to Action: Strengthen global vaccination systems — expand supply chains, rebuild public trust, and accelerate vaccine innovation for diseases with incomplete control.

Source: Our World in Data · Roush & Murphy (2007) | Data accurate as of source brief